

Mining Multi-Dimensional Associations

- ❑ **Single-dimensional** rules (e.g., items are all in “product” dimension)
 - ❑ $\text{buys}(X, \text{“milk”}) \Rightarrow \text{buys}(X, \text{“bread”})$
- ❑ **Multi-dimensional** rules (i.e., items in ≥ 2 dimensions or predicates)
 - ❑ Inter-dimension association rules (*no repeated predicates*)
 - ❑ $\text{age}(X, \text{“18-25”}) \wedge \text{occupation}(X, \text{“student”}) \Rightarrow \text{buys}(X, \text{“coke”})$
 - ❑ Hybrid-dimension association rules (*repeated predicates*)
 - ❑ $\text{age}(X, \text{“18-25”}) \wedge \text{buys}(X, \text{“popcorn”}) \Rightarrow \text{buys}(X, \text{“coke”})$
- ❑ Attributes can be categorical or numerical
 - ❑ Categorical Attributes (e.g., *profession*, *product*: no ordering among values):
Data cube for inter-dimension association
 - ❑ Quantitative Attributes: Numeric, implicit ordering among values—
discretization, clustering, and gradient approaches